

Giona Sala

📍 Nationality: Swiss

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in gionasala

Research experience

- PhD** **Rheinisch-Westfälische Technische Hochschule (RWTH) Aachen**, Aachen, DE
 Doctorate in Theoretical Physics Oct 2023 – present
- Topic: *Gravitational Wave Cosmology*
 - Supervisors: Dr. Sebastian Baum & Prof. Julien Lesgourgues

Education

- MSc** **École Polytechnique Fédérale de Lausanne (EPFL)**, Lausanne, CH
 Master of Science in Physics Sep 2020 – Apr 2023
- Focus on Theoretical Particle Physics
 - Minor in space technologies
- Université de Genève (UniGe)**, Geneva, CH
 Master's thesis Sep 2022 – Apr 2023
- Title: *Implications of Primordial Black Holes physics at high red-shifts*
 - Supervisor: Prof. Antonio Riotto
- BSc** **Eidgenössische Technische Hochschule Zürich (ETHZ)**, Zürich, CH
 Bachelor of Science in Physics Sep 2017 – Sep 2020
- Final project on Hardy's Paradox
- University of Toronto (UofT)**, Toronto, CA
 Exchange Sep 2019 – Dec 2019

Publications

IN G.Sala.5

ID 0009-0001-3716-862X

OR SalaPh

Primary contribution

- Detecting White Dwarf Binary Mergers with Gravitational Waves** Jun 2026
Giona Sala, Chiara Brandenstein, Sebastian Baum, Peter W. Graham
 Phys. Rev. D 113 (2026) 123022 [↗](#)
- Inferring cosmological parameters from galaxy and dark sirens cross-correlation** May 2026
Giona Sala, Alessandro Cuoco, Julien Lesgourgues, Kostantinos-Rafail Revis, Lorenzo Valbusa Dall'Armi, Santiago Casas
 JCAP 05 (2026) 095 [↗](#)

Supporting contribution

- Cosmology beyond standard sirens: cross-correlation of gravitational waves and neutral hydrogen intensity mapping** July 2026
 Matteo Schulz, Ulyana Dupletsa, Andrea Cozzumbo, **Giona Sala**, Tommaso Ronconi, Marta Spinelli, Riccardo Murgia, Simone Mastrogiovanni, Jan Harms, Erik de la Haye, Felicitas Keil
 arXiv:2607.26193 [astro-ph.CO] [↗](#)

Additional relevant experiences

Teaching experience

RWTH,
Teaching assistant

Aachen, DE
Apr 2024 – present

- Supervision:
 - Master's thesis on *Component separation algorithms for the gravitational wave backgrounds*
 - Bachelor's thesis on *Separation of the cosmological and astrophysical gravitational wave background*
- Courses: Theory of Relativity and Cosmology x2; The ingredients of the universe; Theoretical Physics I: Mechanics

EPFL,
Teaching assistant

Lausanne, CH
Sep 2021 – Jan 2023

- Courses: General Physics: Electromagnetism x2, General Physics II, Experimental Physics I, Physics IV

Work experience

Fondazione Istituto Ricerche Solari (IRSOL),
Civil Service

Locarno-Monti, CH
Apr 2023 – Aug 2023

- Solar research technical and scientific collaborator

Scuola Universitaria Professionale della Svizzera Italiana (SUPSI),
Civil Service

Mendrisio, CH
Jun 2022 – Jul 2022

- Renewable energy research collaborator

Extracurricular activities

EPFL Spacecraft Team (EST),
System engineer team leader

Lausanne, CH
Sep 2021 – Jun 2022

- Student association that coordinates the CHESS (Constellation of High Energy Swiss Satellites) space mission

EPFL Rocket Team (ERT),
Payload team leader

Lausanne, CH
Sep 2020 – Oct 2021

- Team of students from EPFL building Bella Lui II 2021 rocket for international competitions

Akademische Raumfahrt Initiative Schweiz (ARIS),
Structure engineer support

Zürich, CH
Feb 2020 – Jul 2020

- The student association for space in the german part of Switzerland, building EULER 2020 rocket

Skills

Programming languages

Python
LaTeX

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C/C++
MATLAB

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Programmes

Mathematica
Excel

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MontePython
CLASS

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Languages

Italian
English

Native
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French
German

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References

Julien Lesgourgues

Prof. Dr.
Institute for Theoretical Particle Physics
and Cosmology
RWTH Aachen
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Sebastian Baum

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Mid Sweden University
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Peter W. Graham

Prof. Dr.
Leinweber Institute for Theoretical Physics
Stanford University
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Antonio Riotto

Prof. Dr.
Department of theoretical physics
Université de Genève
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